Chapter 3.
Determinant & Diagonalization



MSc. Nguyen Cam Nhiem
Email: nhiemnc@fe.edu.vn

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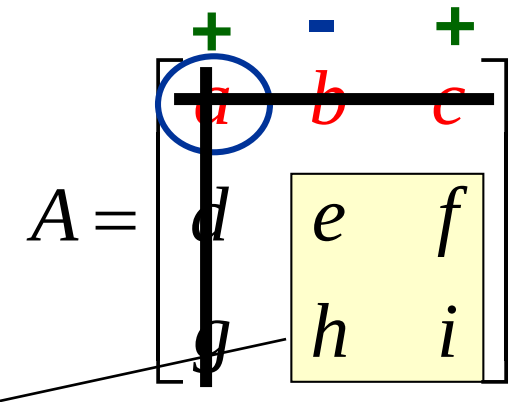
3.1. The Cofactor Expansion

- If A=[a] then the determinant of A, denoted by detA=a
- If A is an 2x2 matrix then

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\det A = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

- If A is an 3x3 matrix then the determinant of A is defined by



$$\det A = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = +a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix}$$

Find detA if

$$A = \begin{bmatrix} 1 & -5 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 3 \end{bmatrix}$$

How to define the determinant of an mxm matrix?

The (i,j)-cofactor

- If A is an mxm matrix then the (i,j)-cofactor of A is defined by

$$c_{ij}(A)=(-1)^{i+j}\det(A_{ij})$$

- A_{ij} is the (m-1)x(m-1) matrix obtained from A by **deleting row i** and **column j** of A
- For example, $c_{23}(A)=(-1)^{2+3}\det(A_{23})=-14$

$$A=\left[\begin{array}{cc|c|c} 1 & 2 & -1 & 5 \\ 0 & 6 & 4 & 3 \\ \hline 0 & 7 & -1 & 0 \\ \hline 0 & 1 & 8 & 2 \end{array}\right] \qquad A_{23}=\begin{bmatrix} 1 & 2 & 5 \\ 0 & 7 & 0 \\ 0 & 1 & 2 \end{bmatrix}$$

Definition

If A is an mxm matrix then the **determinant** of A is defined by

- **detA**=$a_{i1}c_{i1}(A)+a_{i2}c_{i2}(A)+\ldots+a_{im}c_{im}(A)$
- or **detA**= $a_{1j}c_{1j}(A)+a_{2j}c_{2j}(A)+\ldots+a_{mj}c_{mj}(A)$

$$\begin{vmatrix} 1 & 2 & -1 & 5 \\ 0 & 6 & 4 & 0 \\ 0 & 7 & -1 & 0 \\ 0 & 1 & 8 & 2 \end{vmatrix} = 1\begin{vmatrix} 6 & 4 & 0 \\ 7 & -1 & 0 \\ 1 & 8 & 2 \end{vmatrix} = -68$$

Examples

- Find the determinant of the matrices

$$A = \begin{bmatrix} 1 & 2 & 5 \\ 0 & 7 & 0 \\ 1 & 2 & 5 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 & 2 & -1 & 5 \\ 0 & 6 & 4 & 3 \\ 0 & 7 & -1 & 0 \\ 0 & 1 & 8 & 2 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 2 & -1 & 5 \\ 0 & 6 & 4 & 3 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

$$C' = \begin{bmatrix} 1 & 2 & -1 & 5 \\ 0 & 6 & 4 & 3 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & -1 & 0 \end{bmatrix}$$

Properties

- If A has **one row of zeros** then detA=0
- If A is an **triangular matrix** then detA is the **product** of the entries on the **main diagonal**

$$\det C = \begin{vmatrix} 1 & 2 & -1 & 5 \\ 0 & 6 & 4 & 3 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 2 \end{vmatrix} = 1.6.(-1).2 = -12$$

Example

Suppose A is an **triangular matrix**, $a_{11}=4$ and $c_{11}(A)=11$. What is the **product of all entries on the main diagonal**?

A)11 B)44 C)-11 D)-44

Determinants and elementary operations

- If B obtained from A by interchanging two rows (or columns) then **detB=-detA**
- For example,

$$C = \begin{bmatrix} 1 & 2 & -1 & 5 \\ 0 & 6 & 4 & 3 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix} \qquad C' = \begin{bmatrix} 1 & 2 & -1 & 5 \\ 0 & 6 & 4 & 3 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & -1 & 0 \end{bmatrix}$$

detC’=-detC=12

Examples

- If two rows (or columns) of a matrix is the same then the determinant is **zero**

$$C = \begin{bmatrix} 1 & 2 & -1 & 5 \\ 0 & 6 & 4 & 3 \\ 1 & 2 & 5 & 6 \\ 1 & 2 & 5 & 6 \end{bmatrix}$$

Determinants and elementary operations

- If B is the matrix obtained from A by multiplying one row (or column) by a nonzero number **k** then detB=**k**detA
- For example,

$$A' = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 7 & 0 \\ 2 & 4 & 10 \end{bmatrix} \qquad A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 7 & 0 \\ 1 & 2 & 5 \end{bmatrix}$$

detA=35, detA’=2detA=2.35=70

Examples

Find a such that

$$\begin{vmatrix} 4 & 8 & -8 \\ 8 & -4 & 4 \\ 0 & 8 & 4 \end{vmatrix} = a \begin{vmatrix} 1 & 2 & -2 \\ 2 & -1 & 1 \\ 0 & 2 & 1 \end{vmatrix} ?$$

a) 4 b) 12 c) 48 d)64

Examples

- Find the determinants of the matrices

$$A = \begin{bmatrix} 1 & 2 & 5 \\ 0 & 7 & 0 \\ 2 & 4 & 10 \end{bmatrix} \qquad A' = \begin{bmatrix} 1 & 2 & 5 \\ 0 & 7 & 0 \\ 1 & 2 & 5 \end{bmatrix}$$

Determinants and elementary operations

- If B is the matrix obtained from A by adding a multiple of one row (or column) to another row (or column) then **detB=detA**
- For example,

$$A = \begin{bmatrix} 1 & 2 & -3 \\ 0 & 7 & 0 \\ 2 & 4 & -10 \end{bmatrix} \qquad A' = \begin{bmatrix} 1 & 2 & -3 \\ 0 & 7 & 0 \\ 0 & 0 & -4 \end{bmatrix}$$

Examples

- Find the determinants of the matrices

$$A=\begin{bmatrix}1 & a & a+1\\ 2 & b & b+2\\ 3 & c & c+3\end{bmatrix}$$

$$\begin{vmatrix}1 & a & a+1\\ 2 & b & b+2\\ 3 & c & c+3\end{vmatrix} \overset{c_3\to c_3-c_1}{=} \begin{vmatrix}1 & a & a\\ 2 & b & b\\ 3 & c & c\end{vmatrix}=0$$

Examples

Find DetA and DetB:

$$A = \begin{bmatrix} 1 & 2 & -1 & 5 \\ 0 & 6 & 4 & 3 \\ 1 & 2 & 5 & 6 \\ 4 & 8 & 20 & 24 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 2 & -1 & 5 \\ 0 & 6 & 4 & 3 \\ 1 & -3 & 4 & 6 \\ 1 & 2 & 4 & 5 \end{bmatrix}$$

$$\begin{vmatrix} 3 & -1 & 2 \\ 2 & 5 & 1 \\ 0 & 0 & 0 \end{vmatrix} = 0$$

(because the last row consists of zeros)

$$\begin{vmatrix} 3 & -1 & 5 \\ 2 & 8 & 7 \\ 1 & 2 & -1 \end{vmatrix} = - \begin{vmatrix} 5 & -1 & 3 \\ 7 & 8 & 2 \\ -1 & 2 & 1 \end{vmatrix}$$

(because two columns are interchanged)

$$\begin{vmatrix} 8 & 1 & 2 \\ 3 & 0 & 9 \\ 1 & 2 & -1 \end{vmatrix} = 3 \begin{vmatrix} 8 & 1 & 2 \\ 1 & 0 & 3 \\ 1 & 2 & -1 \end{vmatrix}$$

(because the second row of the matrix on the left is 3 times the second row of the matrix on the right)

$$\begin{vmatrix} 2 & 1 & 2 \\ 4 & 0 & 4 \\ 1 & 3 & 1 \end{vmatrix} = 0$$

(because two columns are identical)

$$\begin{vmatrix} 2 & 5 & 2 \\ -1 & 2 & 9 \\ 3 & 1 & 1 \end{vmatrix} = \begin{vmatrix} 0 & 9 & 20 \\ -1 & 2 & 9 \\ 3 & 1 & 1 \end{vmatrix}$$

(because twice the second row of the matrix on the left was added to the first row)

Example

- Find $\det\begin{bmatrix} 6 & 18 \\ -1 & -5 \end{bmatrix}$

and compare with $\det\left(3\begin{bmatrix} 2 & 6 \\ -1 & -5 \end{bmatrix}\right)$?

Theorem

- $\det A = \det(A^T)$
- $\det(AB) = \det A . \det B$
- If A is an nxn matrix then $\det(kA) = k^n \det A$

Note that $\det(A^k) = (\det A)^k$

Example

- Show that if A is an invertible matrix then

$$\det(A^{-1})=1/\det A$$

Example

Let A is a nxn matrix, k is a scalar.
Which of the following are **false**?

(i) $\det(A^T)=\det A$

(ii) $\det(AB)=\det A \det B$

(iii) $\det(A+B)=\det A+\det B$

(iv) $\det(kA)=k\det A$

(v) $\det(kA)=k^n\det A$

Example

Let A and B are nxn matrix. Which statement is **false**?

(i) $\det(A^{-1}BA)=\det B$

(ii) $\det(AB^{-1}A^{-1}B)=1$

(iii) $\det A \det A^{-1}=1$

(iv) $\det A \det A^{T}=1$

(v) $\det(A^{T}B)=\det(B^{T}A)$

Example

$$\det A = \begin{vmatrix} 1 & -1 & -1 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 4 \end{vmatrix}$$

a)detA=12 and $\det(A^{-1})^T=12$

b) detA=12 and $\det(A^{-1})^T=1/12$

c) detA=1/12 and $\det(A^{-1})^T=12$

d) detA=1/12= $\det(A^{-1})T$

Determinant with block

Compute

$$\begin{vmatrix} 1 & 7 & 2 & 6 \\ 5 & -2 & 4 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 3 & 4 \end{vmatrix}$$

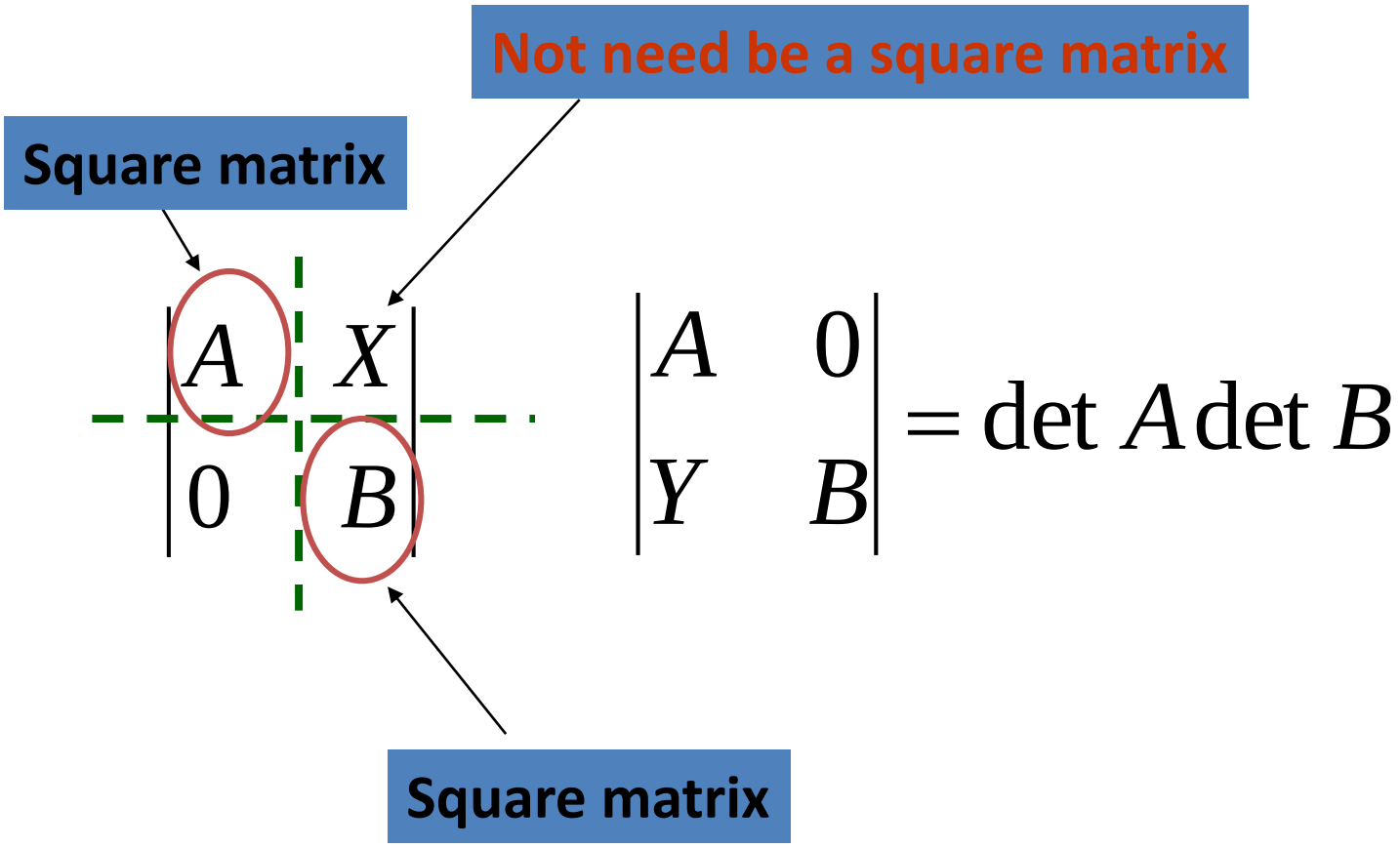
$$\begin{vmatrix} 1 & 7 & 0 & 0 \\ 5 & -2 & 0 & 0 \\ 3 & 1 & 1 & 2 \\ 3 & -2 & 3 & 4 \end{vmatrix}$$

Determinant with block

$$\begin{vmatrix} 1 & 2 & m & a & b \\ 4 & 5 & n & c & d \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 4 & 0 & 6 \\ 0 & 0 & 7 & 0 & 9 \end{vmatrix}$$

$$\begin{vmatrix} 1 & 0 & 0 & a & b \\ 4 & 5 & 6 & c & d \\ 7 & 8 & 9 & e & f \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 3 & 4 \end{vmatrix}$$

Determinant with block



$$\begin{vmatrix} A & X \\ 0 & B \end{vmatrix} = \begin{vmatrix} A & 0 \\ Y & B \end{vmatrix} = \det A \det B$$

Determinant with block

$$\begin{vmatrix} A & 0 \\ Y & B \end{vmatrix} = \det A \det B$$

$$\begin{vmatrix} A & X \\ 0 & B \end{vmatrix} = \det A \det B$$

Determinant with block

Compute

$$\begin{vmatrix} 1 & 7 & 2 & 6 \\ 5 & -2 & 4 & 1 \\ \hline 0 & 0 & 1 & 2 \\ 0 & 0 & 3 & 4 \end{vmatrix}$$

$$\begin{vmatrix} 1 & 7 & 0 & 0 \\ 5 & -2 & 0 & 0 \\ \hline 3 & 1 & 1 & 2 \\ 3 & -2 & 3 & 4 \end{vmatrix}$$

Determinant with block

$$\begin{vmatrix} 1 & 2 & m & a & b \\ 4 & 5 & n & c & d \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 4 & 0 & 6 \\ 0 & 0 & 7 & 0 & 9 \end{vmatrix}$$

$$\begin{vmatrix} 1 & 0 & 0 & a & b \\ 4 & 5 & 6 & c & d \\ 7 & 8 & 9 & e & f \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 3 & 4 \end{vmatrix}$$

3.2.Determinant and Matrix Inverses

- If A is invertible then A→Identity matrix
- If detA ≠0 and A→B by elementary operations then detB ≠0
- A is invertible **iff** detA≠0
- Cramer's Rule
- Give a formula to find A^{-1}

Adjugate matrix

- The adjugate matrix of A is the matrix

$$adjA = \begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & C_{22} & \cdots & C_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ C_{n1} & C_{n2} & \cdots & C_{nn} \end{bmatrix}^T = \begin{bmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & C_{22} & \cdots & C_{n2} \\ \cdots & \cdots & \cdots & \cdots \\ C_{1n} & C_{2n} & \cdots & C_{nn} \end{bmatrix}$$

Find the adjugate of each of the following matrices.

(a) $\begin{bmatrix} 5 & 1 & 3 \\ -1 & 2 & 3 \\ 1 & 4 & 8 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & -1 & 2 \\ 3 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$

Adjugate matrix

Ex: Compute the adjugate of

$$A = \begin{bmatrix} 1 & 3 & -2 \\ 0 & 1 & 5 \\ -2 & -6 & 7 \end{bmatrix}$$

and calculate A(adjA) and (adjA)A.

Theorem of Adjugate Formula

If A is any square matrix, then

- A(adjA)=(detA)**I**
- In particular, if **detA≠0** then A is **invertible** and

$$A^{-1} = \frac{1}{\det A} adjA$$

Adjugate matrix

Ex: Find A^{-1} of

$$A = \begin{bmatrix} 1 & 3 & -2 \\ 0 & 1 & 5 \\ -2 & -6 & 7 \end{bmatrix}$$

Example

Let A be a nxn matrix with detA=3.

Find det(adjA)?

a) 1/3 b)1/2 c)2 d) 3

e) 3^{n-1}

Cramer's Rule

- Cramer's system: system of **n equations** involving **n variables** and has **unique solution**
- So, the **coefficient** matrix is invertible
- A is invertible **iff** detA≠0

Cramer's Rule

- Consider the matrix equation **AX**=B where **A** is an **invertible** square matrix, **X=[x_1 x_2 … x_n]^T**. That means, **AX**=B is a **Cramer's system**. Then the system has **unique solution** given by

$$x_i = \frac{\det A_i}{\det A}$$

?

Cramer's system: AX=B

$$A=\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} \quad X=\begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{bmatrix} \quad B=\begin{bmatrix} b_1 \\ b_2 \\ \dots \\ b_n \end{bmatrix}$$

$$A_2=\begin{bmatrix} a_{11} & \dots & a_{1n} \\ a_{21} & \dots & a_{2n} \\ \dots & \dots & \dots \\ a_{n1} & \dots & a_{nn} \end{bmatrix}$$

$$x_i=\frac{\det A_i}{\det A}$$

Example

Find x_1, given the following system of equations.

$$\begin{aligned} 5x_1 + x_2 - x_3 &= 4 \\ 9x_1 + x_2 - x_3 &= 1 \\ x_1 - x_2 + 5x_3 &= 2 \end{aligned}$$

Solution ▶ Compute the determinants of the coefficient matrix A and the matrix A_1 obtained from it by replacing the first column by the column of constants.

$$\det A = \det\begin{bmatrix} 5 & 1 & -1 \\ 9 & 1 & -1 \\ 1 & -1 & 5 \end{bmatrix} = -16$$

$$\det A_1 = \det\begin{bmatrix} 4 & 1 & -1 \\ 1 & 1 & -1 \\ 2 & -1 & 5 \end{bmatrix} = 12$$

Hence, $x_1 = \dfrac{\det A_1}{\det A} = -\frac{3}{4}$ by Cramer's rule.

- Note that […]

Cramer's rule enabled us to calculate x_1 here without computing x_2 or x_3. This might seem an advantage, but the truth of the matter is that, for large systems of equations, the number of computations needed to find all the variables by the Gaussian algorithm is comparable to the number required to find one of the determinants involved in Cramer's rule, or the matrix of system is not invertible and even the coefficient matrix is not square.

3.3. Diagonalization and Eigenvalues

The following statements are equivalent

- A is **invertible**
- **detA≠0**
- The matrix equation AX=B has unique solution
- The homogeneous system AX=0 has only **trivial solution**
- A→Identity matrix by elementary operations

Example

- Compute A^2,A^3, and A^{2009} if $A = \begin{bmatrix} -3 & 0 \\ 0 & 2 \end{bmatrix}$

$$A^2 = \begin{bmatrix} (-3)^2 & 0 \\ 0 & 2^2 \end{bmatrix}, A^3 = \begin{bmatrix} (-3)^3 & 0 \\ 0 & 2^3 \end{bmatrix}$$

$$A^{2009} = \begin{bmatrix} (-3)^{2009} & 0 \\ 0 & 2^{2009} \end{bmatrix}$$

Diagonal matrices

- An nxn matrix is called diagonal matrix if all its entries off the main diagonal are zeros

$$D = \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & \lambda_n \end{bmatrix} = diag(\lambda_1, \lambda_2, \dots, \lambda_n)$$

- For example

$$diag(3, -2, 1, 4) = \begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

Theorem

- If A is an nxn matrix and P is an invertible nxn matrix then

$$\left(P^{-1}AP\right)^{k} = P^{-1}A^{k}P$$

$$\left(P^{-1}AP\right)^{k} = \left(P^{-1}AP\right)\left(P^{-1}AP\right)\left(P^{-1}AP\right)\ldots\left(P^{-1}AP\right)$$

$$= P^{-1}APP^{-1}APP^{-1}AP\ldots P^{-1}AP = P^{-1}A^{k}P$$

Objective

Find the invertible matrix P such that $P^{-1}AP$ is a diagonal matrix

$$\left(P^{-1}AP\right) = diag\left(\lambda_1, \lambda_2, ..., \lambda_n\right)$$

- Then,

$$P^{-1}A^kP = \left(P^{-1}AP\right)^k = diag\left(\lambda_1{}^k, \lambda_2{}^k, ..., \lambda_n{}^k\right)$$

Example

$$\text{Find } A^{2009} = \left(\begin{bmatrix} 2 & -4 \\ -1 & -1 \end{bmatrix}\right)^{2009} ?$$

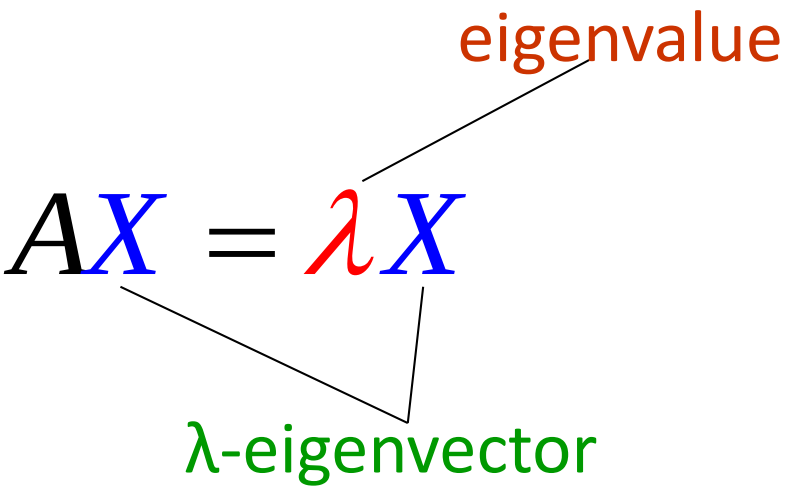
$$P = \begin{bmatrix} 4 & 1 \\ -1 & 1 \end{bmatrix} \qquad P^{-1} = \frac{1}{5}\begin{bmatrix} 1 & -1 \\ 1 & 4 \end{bmatrix}$$

$$P^{-1}AP = \begin{bmatrix} 3 & 0 \\ 0 & -2 \end{bmatrix} \Leftrightarrow \left(P^{-1}AP\right)^{2009} = \left(\begin{bmatrix} 3 & 0 \\ 0 & -2 \end{bmatrix}\right)^{2009}$$

$$\Leftrightarrow P^{-1}A^{2009}P = \begin{bmatrix} 3^{2009} & 0 \\ 0 & (-2)^{2009} \end{bmatrix} \Rightarrow A^{2009} = P\begin{bmatrix} 3^{2009} & 0 \\ 0 & (-2)^{2009} \end{bmatrix}P^{-1}$$

Eigenvalues and eigenvectors

- If A is an nxn matrix, a number λ is called an **eigenvalue** of A if **AX=λX** for some column X≠0
- Such a nonzero column X is called an **eigenvector** of A corresponding to λ, or a λ-eigenvector for short



Example

- Find an eigenvalue and eigenvector for the matrix

$$A = \begin{bmatrix} 3 & 5 \\ 1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 5 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 5 \\ 1 \end{bmatrix} = 4\begin{bmatrix} 5 \\ 1 \end{bmatrix}$$

eigenvalue

eigenvector

$$\begin{bmatrix} 3 & 5 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 1 \\ -1 \end{bmatrix} = -2\begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Characteristic Polynomial

- The characteristic polynomial of an nxn matrix A is defined by

$$c_A(x)=\det(xI-A)$$

- For example, if $A=\begin{bmatrix} 3 & 5 \\ 1 & -1 \end{bmatrix}$

$$c_A(x)=\det(xI-A)=\det\left(\begin{bmatrix} x-3 & -5 \\ -1 & x+1 \end{bmatrix}\right)$$

$$=(x-3)(x+1)-5=x^2-2x-8$$

Theorem

- *Let A be n x n matrix.*

1. *The eigenvalues λ of A are the roots of the characteristic polynomial $c_A(x)$ of A.*
2. *The λ-eigenvectors* **x** *are the nonzero solutions to the homogeneous system*

$$(\lambda I - A)\mathbf{x} = \mathbf{0}$$

of linear equations with λI - A as coefficient matrix.

Example

Find the eigenvalues and eigenvectors of the matrix

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$$

The characteristic polynomial of A is

$$c_A(x) = \begin{vmatrix} x-1 & -1 \\ -2 & x-2 \end{vmatrix} = (x-1)(x-2) - 2$$

$$= x(x-3)$$

$$c_A(x) = 0 \Leftrightarrow x(x-3) = 0$$

- $\lambda_1 = 0$ we solve the equation

$$\begin{bmatrix} -1 & -1 \\ -2 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} -1 & -1 \\ 0 & 0 \end{bmatrix}$$

The solution is

$$X = t\begin{bmatrix} -1 \\ 1 \end{bmatrix}$$ t is arbitrary

The eigenvectors of A corresponding to $\lambda_1=0$ is

$$X = t\begin{bmatrix} -1 \\ 1 \end{bmatrix}$$ $t \neq 0$

- $\lambda_2 = 3$ we solve the equation

$$\begin{bmatrix} 2 & -1 \\ -2 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & -1 \\ 0 & 0 \end{bmatrix}$$

The solution is

$$X = t\begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix}$$ t is arbitrary

The eigenvectors of A corresponding to $\lambda_2 = 3$ is

$$X = t\begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix}$$ t≠0

When is A diagonalizable?

Theorem

A is diagonalizable iff every eigenvalue λ of multiplicity m

yields exactly m basic eigenvectors, that is,

iff the general solution of the system $(\lambda I - A)X = 0$

has exactly m parameters.

EXAMPLE 9

Diagonalize the matrix $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$.

Solution ▶ To compute the characteristic polynomial of A first add rows 2 and 3 of $xI - A$ to row 1:

$$c_A(x) = \det\begin{bmatrix} x & -1 & -1 \\ -1 & x & -1 \\ -1 & -1 & x \end{bmatrix} = \det\begin{bmatrix} x-2 & x-2 & x-2 \\ -1 & x & -1 \\ -1 & -1 & x \end{bmatrix}$$

$$= \det\begin{bmatrix} x-2 & 0 & 0 \\ -1 & x+1 & 0 \\ -1 & 0 & x+1 \end{bmatrix} = (x-2)(x+1)^2$$

Hence the eigenvalues are $\lambda_1 = 2$ and $\lambda_2 = -1$, with λ_2 repeated twice (we say that λ_2 has *multiplicity* two). However, A *is* diagonalizable. For $\lambda_1 = 2$, the system of equations $(\lambda_1 I - A)\mathbf{x} = \mathbf{0}$ has general solution $\mathbf{x} = t\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ as the reader can verify, so a basic λ_1-eigenvector is $\mathbf{x}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$.

Turning to the repeated eigenvalue $\lambda_2 = -1$, we must solve $(\lambda_2 I - A)\mathbf{x} = \mathbf{0}$. By gaussian elimination, the general solution is $\mathbf{x} = s\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + t\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ where s and t are arbitrary. Hence the gaussian algorithm produces *two* basic λ_2-eigenvectors $\mathbf{x}_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$ and $\mathbf{y}_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$. If we take $P = [\mathbf{x}_1 \ \mathbf{x}_2 \ \mathbf{y}_2] = \begin{bmatrix} 1 & -1 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$ we find that P is invertible. Hence $P^{-1}AP = \text{diag}(2, -1, -1)$ by Theorem 4.

P135: 1-11

P148: 1-10, 20

P166: 1, 8